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Computer Science Research Report

# MetaShift-Bench: A Metadata-Anchored Audit Benchmark for PM2.5 Method-Transition Discontinuities

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## Abstract

Regulatory monitoring records can exhibit discontinuities caused by changing environmental conditions, data processing, or a reported measurement method. An isolated time series cannot distinguish a station-specific discontinuity from a regional pattern. This report presents MetaShift-Bench, a reproducible metadata-anchored audit benchmark for reported United States Environmental Protection Agency (EPA) Air Quality System (AQS) PM2.5 Method Code transitions. A Method Code transition is treated as a reproducible event anchor, not as proof of physical instrument replacement, fault, or causal measurement bias. The protocol constructs pre-transition geographically constrained counterfactuals, compares cross-site and single-series methods under known synthetic perturbations, and preserves every eligible real event with its diagnostics or explicit failure reason.

The frozen evidence inventory contains 2,424,793 canonical daily observations from 1,689 monitor time series and 563 persistent anchors. Only 238 anchors retain at least three distinct physical donors; 228 receive complete common-method comparisons, while 325 are donor-insufficient and 10 have estimator input-window failures. The stable-regime benchmark uses 146 physical target sites split into 66 calibration and 80 evaluation sites with disjoint complete target-plus-donor footprints. Standard synthetic control has aggregate local-effect MAE 0.09983, AUPRC 0.91476, and macro-F1 0.81641. Cross-validated MetaShift has lower point-estimate MAE (0.09742) but its paired difference versus standard synthetic control is -0.00240 (95% CI [-0.00772, 0.00347]). Thus the evidence does not support a general MetaShift superiority claim. The contribution is instead a transparent benchmark, complete event audit, layered diagnostics, and a reproducible account of applicability limits.

**Keywords:** air-quality monitoring; PM2.5; metadata anchors; synthetic control; change-point attribution; reproducible audit.

**Frozen-evidence statement.** All quantitative findings in this report are generated from the v0.3.2-evidence-final evidence release at commit 57d678ecabebff724d898abe626c9ef80538775b; see the [official frozen evidence release](#). The full case-manifest SHA-256 is recorded in the frozen evidence configuration and reproducibility table. This document does not modify the frozen study protocol or reopen model tuning.

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# 1 Introduction

Air-quality measurements support public communication, regulatory compliance, trend analysis, exposure assessment, and health research. A visible break in a monitor’s PM2.5 series can have multiple explanations. It may reflect meteorology, wildfire smoke, local emissions, siting, operational changes, data processing, or a change in the reported measurement method. These explanations cannot be separated by a numerical change-point alone.

The AQS data model makes it possible to formulate a narrower, auditable question. Its daily records expose a reported Method Code, a Method Name, and a Parameter Occurrence Code (POC) for each series [11, 12, 15]. A Method Code transition can therefore define a reproducible candidate date before inspecting local residual effects. It does not reveal a physical instrument log: a POC is not a universal physical-instrument identifier, and the code transition may have several non-hardware explanations.

MetaShift-Bench asks whether the target-minus-reference residual near a reported transition is unusual relative to stable dates and reference sites. The goal is an evidence audit, not an automatic correction of historical concentrations. Cross-site counterfactual methods can help isolate station-specific departures, but they rely on pre-event agreement and geographically plausible reference series. Synthetic control and difference-in-differences supply useful counterfactual and placebo precedents [1, 3]; they do not turn a monitoring metadata record into a causal intervention.

The report addresses five predeclared questions:

1. **RQ1:** How do cross-site and single-series methods behave under known local and matched regional perturbations in stable monitoring windows?
2. **RQ2:** Does either frozen MetaShift variant establish a confidence-supported aggregate advantage over standard synthetic control?
3. **RQ3:** What fraction of the complete metadata-anchor universe supports a common cross-site comparison, and what do its placebo and sensitivity diagnostics show?
4. **RQ4:** How well do the reported interval constructions cover known synthetic effects, and what uncertainty remains uncalibrated?
5. **RQ5:** What contextual support is available from same-site POC records, official documents, and a separate PM2.5 parameter pipeline?

The central contribution is a measurement-method transition *audit benchmark*: public source provenance, a complete eligible-event ledger, a distinct-physical-donor rule, synthetic truth, placebos, sensitivity analyses, and explicit abstention. No taxonomy-stratified analysis is reported because the frozen Method Code taxonomy remains pending independent student and teacher review. The benchmark does not claim that a Method Code change proves a device replacement, a measurement failure, or a causal measurement bias.

## 2 Background and related work

### 2.1 Monitoring comparability and public metadata

Monitoring comparability is an empirical problem rather than a label inferred from a single administrative field. EPA publishes AQS file formats, a Method Code reference, and continuous-monitor comparability resources [11, 12, 14]. Collocated Federal Reference Method and Federal Equivalent Method studies show why configuration-specific comparison is important [7]. They do not, however, supply a dated physical-change ground truth for every historical AQS Method Code transition.

Low-cost PM2.5 sensor studies similarly motivate careful calibration and reference comparison

[2, 4, 5]. Their calibration settings, instruments, and outcomes differ from a retrospective audit of reported regulatory-monitor metadata. MetaShift-Bench uses these works as measurement-quality context, not as evidence that an AQS transition has a particular physical cause.

## 2.2 Reference series, counterfactuals, and change points

Single-series change-point methods such as PELT identify changes in a sequence efficiently [8]. Environmental analyses can combine change-point tests with meteorological normalization [6]. Such methods remain valuable baselines, but a single series cannot determine whether a discontinuity is local to the monitor or shared across a region.

Pairwise homogenization of climate records is a closer design analogue: reference stations and station histories can expose inhomogeneity while leaving causal interpretation conditional on assumptions [9]. Synthetic control provides donor-weighted counterfactuals and in-space placebo logic [1]; difference-in-differences formalizes stronger treatment identification assumptions [3]. This study adapts those ideas as diagnostics, not as proof that the causal assumptions of a policy intervention apply to a reported monitoring-method transition.

## 2.3 Contribution boundary

MetaShift-Bench integrates established ingredients into one constrained workflow: AQS Method Code metadata defines anchors; distinct physical sites define geographic donors; stable windows receive known synthetic perturbations; and every eligible real anchor receives either a result or an explicit exclusion. We are not aware of a prior benchmark that jointly uses reported AQS measurement-method transitions as reproducible metadata anchors, cross-site diagnostic controls, stable-window synthetic perturbations, and complete eligible-event accounting. This is a scoped literature statement, not an assertion of being first in a broader field.

# 3 Data and event construction

## 3.1 Primary public-data corpus

The primary corpus comprises EPA AirData daily archives for parameter 88101, PM2.5 local conditions, for 2019–2025 [11, 13]. The canonical signal is the `Arithmetic Mean` for 24-HR BLK AVG observations. The predeclared cleaning rule retains nonempty Method Code records with observation percent at least 75 and an Event Type other than `Excluded`. It prevents alternative daily summaries for the same monitor-day from being treated as independent observations.

The source files, archive size, checksum, selected CSV member, and retained row count are preserved in the frozen data manifest. The daily data are public, but raw downloads and API responses are not included in this manuscript repository or its paper assets. That separation keeps the report reproducible without placing raw files or local credentials under version control.

## 3.2 Metadata anchors and distinct physical donors

An anchor is a persistent reported Method Code transition with at least 60 calendar days on each side, at least 45 retained observations in each adjacent method run, and no more than a seven-day transition gap. An eligible geographic donor is at a different physical site, within the predeclared distance limit, method-stable in the target window, sufficiently paired in the pre-event period, and sufficiently correlated before the anchor.

**Table 1:** Frozen v0.3.2 data and audit inventory.

| Quantity                                             | Count     |
|------------------------------------------------------|-----------|
| Canonical daily PM2.5 records                        | 2,424,793 |
| Monitor time series                                  | 1,689     |
| Persistent Method Code anchors                       | 563       |
| Anchors with at least one distinct physical donor    | 394       |
| Anchors with at least three distinct physical donors | 238       |
| Complete common-method comparisons                   | 228       |
| Donor-insufficient anchors                           | 325       |
| Estimator input-window failures                      | 10        |

*Note.* The physical donor identifier is State Code + County Code + Site Number; at most one POC is retained per donor site.

Physical donor identity is State Code + County Code + Site Number. A target’s candidate donor list can contain multiple POCs for one physical site, but the ranked donor rule retains at most one: highest pre-event correlation, then shorter distance, then more paired days, then POC as a deterministic tie-breaker. This rule remediates an earlier superseded inventory in which multiple POCs could be treated as independent geographic donors. Same-site alternate POCs are excluded from geographic donor construction and reserved for contextual consistency analysis.

### 3.3 Audit disposition

Every anchor is retained in the event audit. A complete comparison means that the required common cross-site inputs exist; donor insufficiency and input-window failure are explicit machine-readable outcomes rather than omitted rows. This complete accounting is necessary because a result on only the well-matched subset would overstate the scope of cross-site inference.

## 4 Methods

### 4.1 Counterfactual residual

For target monitor  $i$ , date  $t$ , and selected geographic donors  $j \in N_i$ , the daily concentration  $y_{i,t}$  is transformed as

$$z_{i,t} = \log(1 + \max(y_{i,t}, 0)).$$

For a donor-weighted counterfactual, the residual is

$$r_{i,t} = z_{i,t} - \sum_{j \in N_i} w_{ij} z_{j,t} - \text{median}_{s \in T_{\text{cal}}} \left( z_{i,s} - \sum_{j \in N_i} w_{ij} z_{j,s} \right).$$

Weights are nonnegative and sum to one. They are fitted using only the pre-anchor calibration interval, ending before a buffer immediately preceding the anchor. The primary effect is the difference between post- and pre-anchor median residuals in symmetric windows. Raw-unit effects are retained as an interpretation aid, not as interchangeable causal estimands.

Fixed-prior MetaShift uses a reliability preference based on pre-event correlation and distance. Cross-validated MetaShift selects among frozen pre-event settings only. Standard synthetic control fits its donor weights without the MetaShift reliability prior; nearest-neighbor difference-in-differences uses the closest qualified donor reference. Before-after median, Bayesian mean shift, CUSUM, rolling-MAD, and PELT provide single-series comparators.

## 4.2 Evidence diagnostics

The real-event audit combines several diagnostics without assigning a physical cause:

1. pre-event counterfactual fit;
2. fixed-weight moving-block intervals;
3. selection-aware nested moving-block intervals that re-evaluate donor correlation eligibility, donor selection, and weight fitting inside the observed candidate pool;
4. stable post-transition time placebos and exploratory Benjamini–Hochberg-adjusted screening quantities;
5. donor-as-treated and date-resampling placebos; and
6. leave-one-donor-out refits.

An anchor is labelled a supported candidate discontinuity only when all predeclared available evidence conditions hold. An event with available but failed conditions is not supported by the available evidence; an event without a necessary comparative or diagnostic input is inconclusive. These are observational audit tiers. They are not labels for instrument failures, confirmed bias, or true pollution corrections.

## 4.3 Inference boundary

The nested procedure captures some time-resampling, donor-reselection, and weight-refitting variation within the observed candidate pool. It does not capture uncertainty in source metadata, site geography, or initial candidate- pool construction. Accordingly, even when an interval excludes zero it is a diagnostic statement about the protocol-defined residual, not a calibrated confidence interval for a physical measurement bias.

# 5 Experimental design

## 5.1 Stable-regime synthetic benchmark

Real metadata anchors have no physical-bias ground truth. Known-effect evaluation therefore uses only stable target and donor regimes that are separated from nearby reported transitions. The frozen `stable_full_v2` manifest contains 146 physical target sites. Whole connected components of the complete target-plus-donor physical input graph were assigned to the 66-site calibration subset or the 80-site held-out evaluation subset, so no physical input site crosses the split.

The benchmark injects target-only additive, proportional, gradual-drift, temporary-step, and variance changes, together with matched target-and-donor regional variants. Each perturbation variant contains 400 held-out evaluation samples. The effect shape, magnitudes, seeds, candidate rule, model settings, and metrics were frozen before evaluation. The viewed evaluation partition is not used for model, threshold, donor-weight, gate, or taxonomy adjustment.

## 5.2 Real-anchor audit and sensitivity

The real-anchor workflow uses the same distinct-physical-donor construction, pre-event fitting boundaries, and effect window across methods. It records complete comparisons and explicit non-complete dispositions for all 563 anchors. Predeclared sensitivity analyses vary screening rules, effect windows, reporting scale, and evidence-tier thresholds one factor at a time. Their purpose is to expose dependence and availability boundaries, not to select a more favorable specification.

**Table 2:** Independent stable-regime synthetic evaluation.

| Method               | MAE     | AUPRC   | Macro-F1 | Regional FPR |
|----------------------|---------|---------|----------|--------------|
| Standard SC          | 0.09983 | 0.91476 | 0.81641  | 0.140        |
| MetaShift fixed      | 0.10344 | 0.89832 | 0.79515  | 0.150        |
| MetaShift CV         | 0.09742 | 0.90178 | 0.80916  | 0.135        |
| Nearest-neighbor DiD | 0.11150 | 0.88443 | 0.78852  | 0.075        |

*Note.* Metrics are aggregates over the frozen `stable_full_v2` evaluation set. Lower MAE and regional false-positive rate are preferable; higher AUPRC and macro-F1 are preferable.

### 5.3 Interval coverage protocol

Fixed-weight interval coverage is evaluated on the held-out synthetic effects. The confidence level, block length, perturbation types, and repetitions are fixed in the frozen configuration. The separate full selection-aware synthetic coverage protocol was frozen before outcomes. Its feasibility assessment records that full donor re-screening and weight refitting at the specified nested scale is infeasible within the submission deadline. The report therefore does not convert real-event selection-aware intervals into coverage-calibrated physical-bias intervals.

### 5.4 External context and reproducibility

Same-site alternate-POC records, narrow hourly POC windows, QA availability, and targeted official-document review supply contextual checks with known limitations. A separate parameter-88502 pipeline uses its own scan and audit; it is never pooled with 88101. The frozen release records the exact source commit, locked Python environment, source manifests, machine-readable checks, and a two-environment comparison of designated core artifact hashes.

## 6 Results

### 6.1 RQ1: Stable-window synthetic comparison

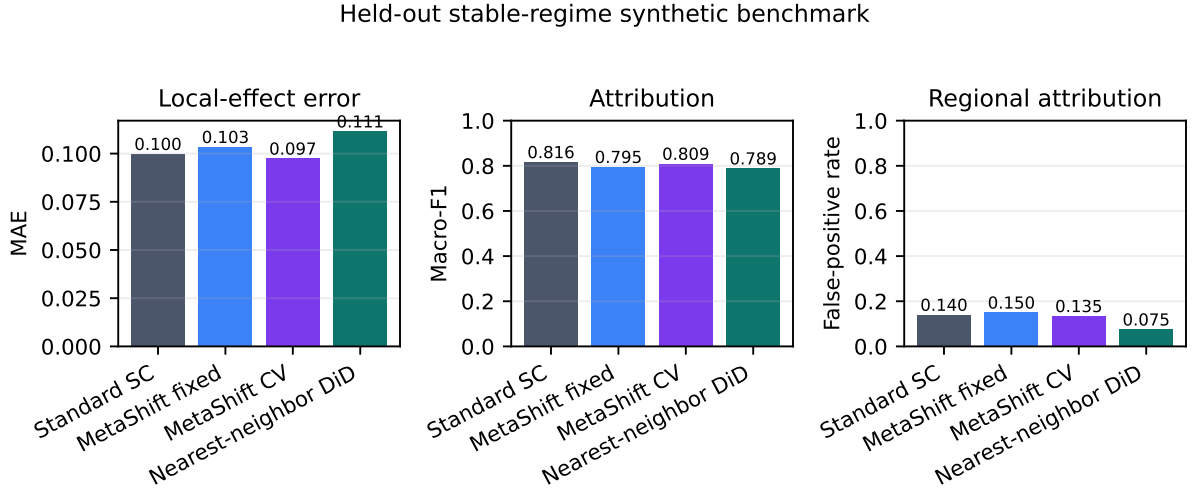
**Answer.** The synthetic benchmark differentiates cross-site methods, but no method is uniformly best across all metrics and perturbation families. Table 2 reports the held-out aggregate results, and Figure 1 visualizes the primary trade-offs.

Standard synthetic control has stronger aggregate attribution ranking and macro-F1 than both MetaShift variants, while cross-validated MetaShift has the smallest aggregate local-effect MAE. Nearest-neighbor difference-in-differences has the lowest regional false-positive rate among these four methods but weaker effect error and attribution metrics. These are benchmark-specific trade-offs, not a ranking for every monitoring network.

### 6.2 RQ2: Bound on a MetaShift superiority claim

**Answer.** Neither MetaShift variant has a confidence-supported aggregate MAE advantage over standard synthetic control. The paired bootstrap intervals in Table 3 cross zero for both frozen variants.

The main and ablation experiments share 7300 standard synthetic-control records exactly to the configured numerical tolerance. Table 4 therefore compares reliability-prior variants without changing the standard-control reference rows. The ablations do not establish a stable

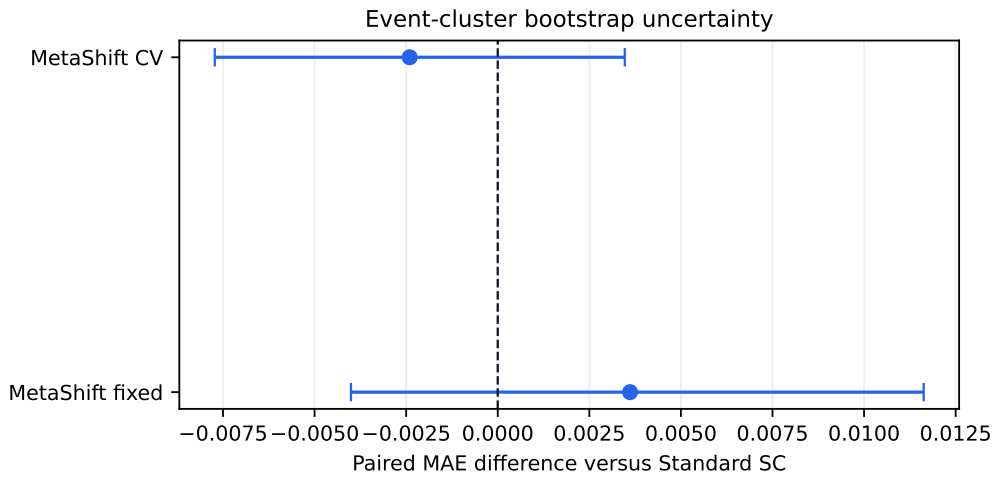


**Figure 1:** Held-out stable-regime synthetic metrics generated from the frozen stable synthetic metrics table.

**Table 3:** Paired event-cluster bootstrap for local-effect MAE differences.

| Comparison                        | Difference | 95% CI              | Clusters |
|-----------------------------------|------------|---------------------|----------|
| MetaShift fixed minus Standard SC | 0.00361    | [-0.00401, 0.01163] | 80       |
| MetaShift CV minus Standard SC    | -0.00240   | [-0.00772, 0.00347] | 80       |

*Note.* Difference is MetaShift minus standard synthetic control; negative values favor MetaShift on MAE. Both intervals include zero.



**Figure 2:** Paired event-cluster bootstrap intervals for MetaShift minus standard synthetic-control local-effect MAE. The dashed line is no difference.

**Table 4:** Reliability-prior and regularization ablations on shared synthetic inputs.

| Variant              | MAE     | Macro-F1 | Regional FPR |
|----------------------|---------|----------|--------------|
| Standard SC          | 0.09983 | 0.81641  | 0.140        |
| No ridge penalty     | 0.10289 | 0.80127  | 0.138        |
| Ridge = 0.01         | 0.10304 | 0.80263  | 0.130        |
| Add coverage         | 0.10332 | 0.79862  | 0.135        |
| MetaShift full prior | 0.10344 | 0.79515  | 0.150        |
| No graph-prior       | 0.10350 | 0.80167  | 0.133        |
| No correlation term  | 0.10362 | 0.79858  | 0.133        |
| No distance term     | 0.10370 | 0.80588  | 0.088        |
| Uniform prior        | 0.10378 | 0.80521  | 0.112        |
| Ridge = 1.0          | 0.10500 | 0.80285  | 0.097        |
| Direct reliability   | 0.10543 | 0.80574  | 0.105        |

*Note.* The common standard-synthetic-control records align exactly across the main and ablation experiments to tolerance  $10^{-10}$ .

**Table 5:** Complete 88101 event audit and real-event diagnostics.

| Audit disposition                         |  |  | Anchors |
|-------------------------------------------|--|--|---------|
| Complete common-method comparison         |  |  | 228     |
| Fewer than three distinct physical donors |  |  | 325     |
| Estimator input-window failure            |  |  | 10      |
| Total                                     |  |  | 563     |

| Method               | Events | Fixed CI excludes 0 | Median log effect |
|----------------------|--------|---------------------|-------------------|
| MetaShift fixed      | 228    | 159                 | -0.08704          |
| Standard SC          | 228    | 145                 | -0.07078          |
| Nearest-neighbor DiD | 228    | 134                 | -0.07490          |

*Note.* Nested selection-aware intervals are available for 227/228 complete events; 153 exclude zero. Their mean width is 0.19057 log units versus 0.17863 for fixed-weight MetaShift intervals. Leave-one-donor-out refits complete for 227 events, with 202 retaining direction under every donor removal. All intervals are diagnostic, not calibrated physical-bias confidence intervals.

overall MetaShift advantage; this negative result closes the model-superiority route for the frozen benchmark.

### 6.3 RQ3: Complete real-anchor audit

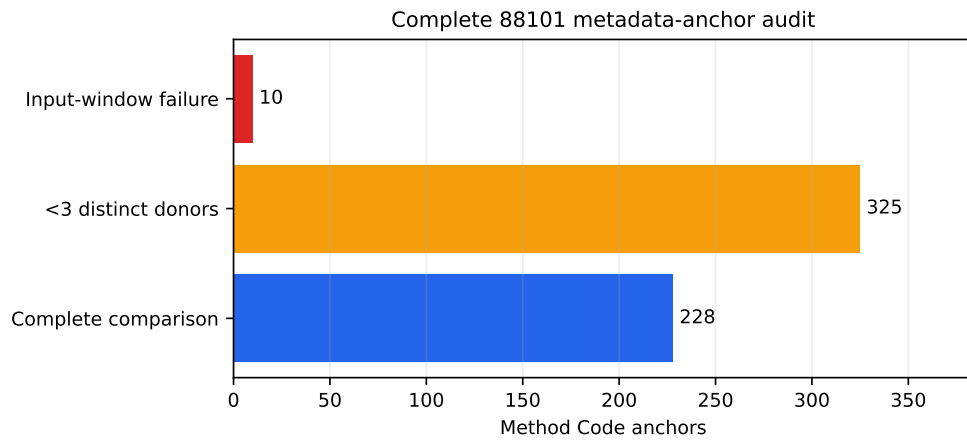
**Answer.** Only 228 of 563 anchors support a complete common cross-site comparison under the distinct-physical-donor rule. The remaining 325 donor-insufficient and 10 input-window events are retained as applicability boundaries rather than removed.

The exploratory evidence synthesis assigns 34 supported candidates, 122 not-supported events, and 407 inconclusive events. The supported label means that predeclared diagnostic conditions were jointly met; it does not confirm a physical failure, replacement, or causal bias.

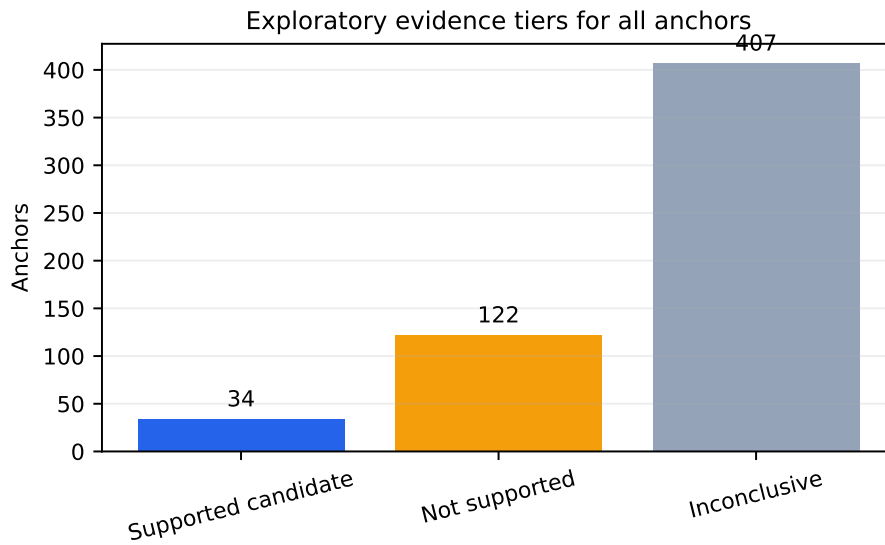
Time-placebo coverage is available for 157 complete events with at least 50 dates, including 128 with 100 dates. The donor-as-treated table contains 866 rows, and the 200-resample date comparison has upper-tail probability 0.00498. These are diagnostic falsifications rather than causal probabilities.

### 6.4 RQ4: Interval coverage and sensitivity

**Answer.** The two available fixed-protocol interval constructions do not support a generic calibration claim. At nominal 95%, fixed-weight conditional block-bootstrap coverage ranges



**Figure 3:** Complete disposition of all 88101 metadata anchors.

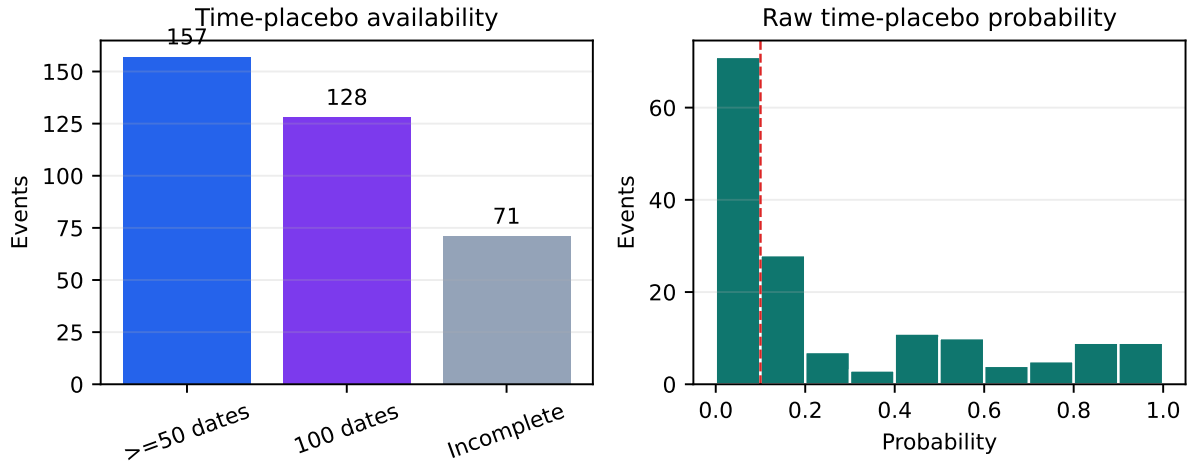


**Figure 4:** Exploratory evidence tiers for the full anchor inventory.

**Table 6:** Placebo, external-document, and same-site POC evidence boundaries.

| Diagnostic                                          | Value |
|-----------------------------------------------------|-------|
| Complete time-placebo events with at least 50 dates | 157   |
| Complete time-placebo events with 100 dates         | 128   |
| Raw time-placebo probabilities at most 0.10         | 71    |
| BH q values at most 0.10                            | 40    |
| Donor-as-treated placebo records                    | 866   |
| Date-resampling permutations                        | 200   |
| Dated site-specific external confirmations          | 0/20  |
| Usable hourly same-site POC comparisons             | 9/11  |
| Daily/hourly POC direction agreements               | 8/9   |
| Standard SC 90th-percentile risk-gate retention     | 75/80 |

*Note.* Time and donor placebos are diagnostic falsifications. Same-site POC and document review provide contextual evidence only, not instrument ground truth.



**Figure 5:** Time-placebo availability and raw within-event probability distribution. The vertical line marks the exploratory 0.10 screening threshold.

**Table 7:** Held-out synthetic interval coverage by method.

| Method               | Fixed 95% coverage | Fixed mean width | Conformal 90% coverage | Conformal mean width |
|----------------------|--------------------|------------------|------------------------|----------------------|
| Standard SC          | 67.281%            | 0.265            | 98.8125%               | 0.396                |
| MetaShift fixed      | 63.875%            | 0.270            | 99.4375%               | 0.471                |
| MetaShift CV         | 65.406%            | 0.265            | 98.9375%               | 0.437                |
| Nearest-neighbor DiD | 66.656%            | 0.287            | 99.5625%               | 0.565                |

*Note.* Each method has 3,200 evaluation effect instances. Fixed-weight intervals under-cover and split-conformal intervals over-cover under this frozen synthetic design.

from 63.875% to 67.281% over 3,200 held-out effect instances per method. At nominal 90%, split-conformal coverage ranges from 98.8125% to 99.5625% and is highly conservative.

Selection-aware nested intervals exist for 227 complete real events and exclude zero for 153. Each requested 1,000 repetitions and retained at least 521 valid repetitions. This result does not validate their coverage: the frozen full selection-aware synthetic coverage protocol is infeasible within the deadline. The intervals are therefore reported as conditional diagnostics.

Table 8 reports predeclared effect-window and reporting-scale sensitivity; Table 9 reports one-factor screening availability. The geographic donor radius visibly changes how many anchors can be analyzed, so these settings are documented rather than optimized after viewing results.

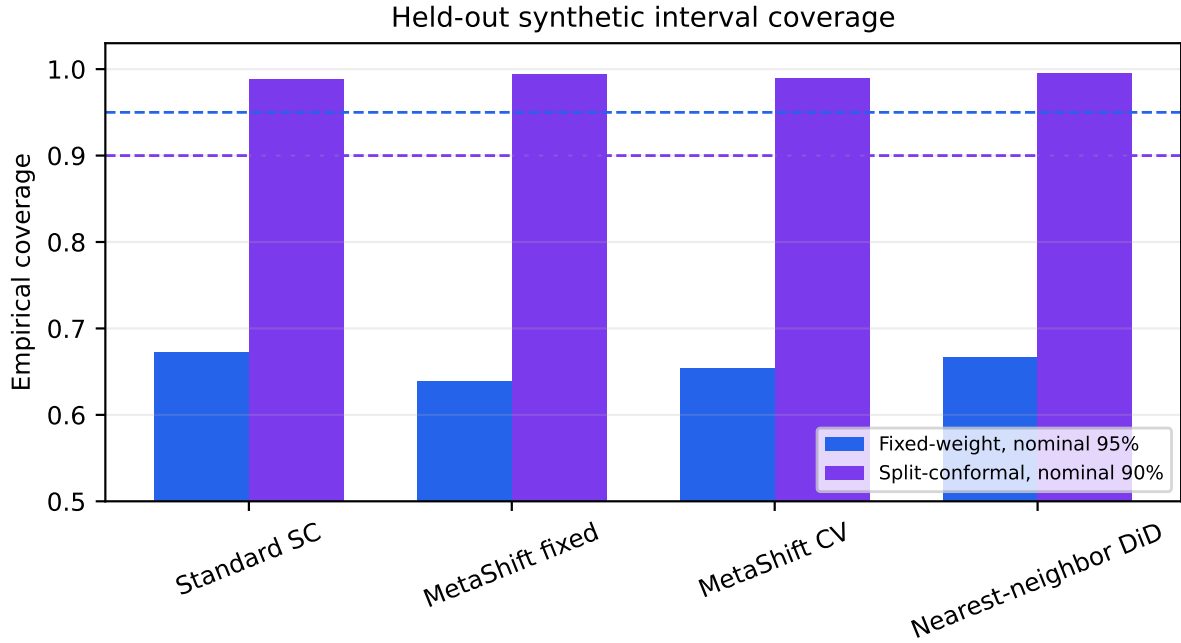
**Table 8:** Predeclared effect-window and reporting-scale sensitivity.

| MetaShift window | Complete events | Median log effect | Sign agreement with 60 days |
|------------------|-----------------|-------------------|-----------------------------|
| 45 days          | 224             | -0.08769          | 93.3%                       |
| 60 days          | 228             | -0.08704          | reference                   |
| 90 days          | 196             | -0.08064          | 93.9%                       |

| Method               | Log/raw sign agreement | Spearman $\rho$ |
|----------------------|------------------------|-----------------|
| MetaShift fixed      | 95.2%                  | 0.829           |
| Standard SC          | 92.5%                  | 0.868           |
| Nearest-neighbor DiD | 92.1%                  | 0.843           |

*Note.* Window sensitivity adds a full method-stability check for every target and donor window. Reporting-scale agreement is a concordance diagnostic and does not equate raw and log causal estimands.

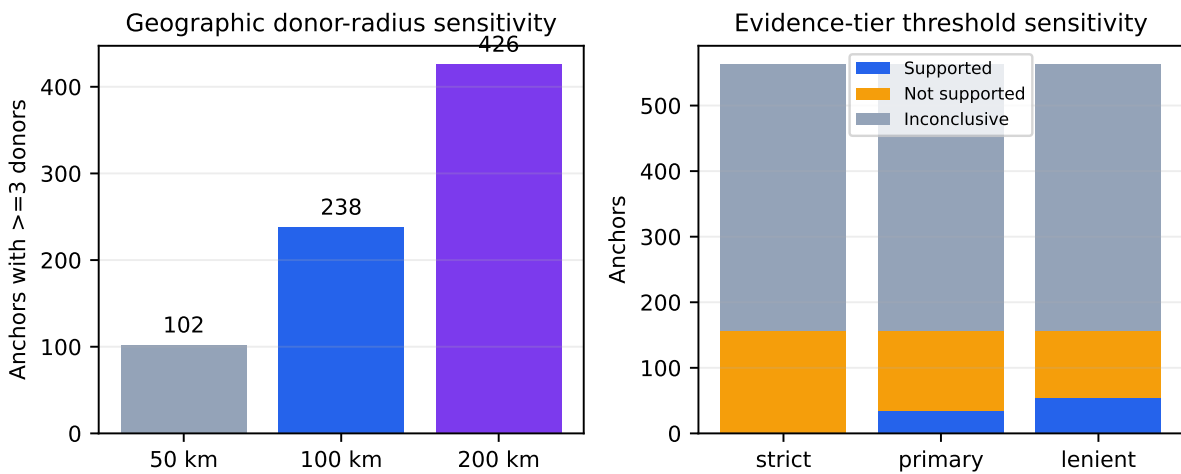


**Figure 6:** Empirical held-out synthetic coverage. Dashed lines show nominal coverage levels for each construction.

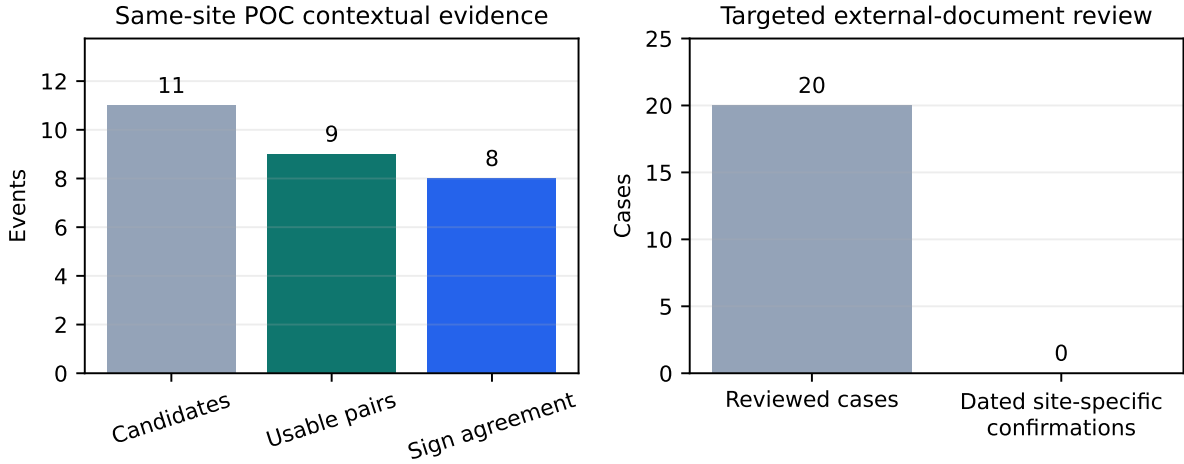
**Table 9:** One-factor screening sensitivity with three required distinct donors.

| Setting                      | Eligible before donor rule | With at least 3 donors |
|------------------------------|----------------------------|------------------------|
| Primary (100 km)             | 563                        | 238                    |
| Coverage 70%                 | 563                        | 243                    |
| Coverage 80%                 | 563                        | 232                    |
| Stable window 45 days        | 572                        | 252                    |
| Stable window 90 days        | 543                        | 214                    |
| Transition gap 3 days        | 512                        | 212                    |
| Transition gap 14 days       | 589                        | 254                    |
| Donor radius 50 km           | 563                        | 102                    |
| Donor radius 200 km          | 563                        | 426                    |
| Correlation $\rho \geq 0.50$ | 563                        | 241                    |
| Correlation $\rho \geq 0.70$ | 563                        | 230                    |

*Note.* One predeclared setting changes at a time. These counts describe audit availability, not detected-effect frequency.



**Figure 7:** Predeclared geographic screening and evidence-tier threshold sensitivity. These describe availability and exploratory tiers, not physical causality.



**Figure 8:** Availability of same-site POC and targeted official-document context.

## 6.5 RQ5: External context and transfer boundary

**Answer.** External evidence supplies limited contextual support, not ground truth. Of 11 same-site POC candidates, 9 have usable paired hourly windows and 8 agree in daily/hourly direction. The targeted official-document review found 0/20 dated site-specific confirmations. A missing public notice is unavailable evidence, not evidence that no physical change occurred.

The independent 88502 pipeline has 34 eligible metadata anchors and only 3 complete common-method comparisons. It demonstrates separate-pipeline feasibility but is too sparse to support broad generalization. EPA documentation about network data alignment provides useful method context [10]; it is not a dated site-specific confirmation for the reviewed anchors.

## 7 Discussion

MetaShift-Bench makes a weakly labelled question auditable rather than pretending that it is fully identified. The critical design choice is to retain metadata anchors, availability constraints, donor rules, algorithm comparisons, and failures in the same evidence chain. A result from a well-matched target is useful only when its selection requirements, reference composition, uncertainty scope, and external evidence limitations remain visible.

The synthetic experiment is the source of known truth in this project. It shows that cross-site counterfactual construction can separate some local and matched regional patterns better than single-series baselines, while also showing that MetaShift does not achieve a stable aggregate advantage over standard synthetic control. Reporting that negative result prevents the benchmark from being reinterpreted as a method-promotion exercise.

The real-event audit has a complementary role. It asks which metadata anchors can be evaluated under transparent, predeclared inputs and which cannot. A complete residual comparison is not evidence of a physical measurement error. Conversely, a donor-insufficient event is not evidence that no change occurred; it is an abstention because the protocol cannot build a qualified geographic reference. This distinction makes the full ledger more informative than a curated set of striking examples.

The evidence tiers are best understood as a prioritization tool for future human review. A supported candidate has a coherent pattern across the predefined diagnostics, but contextual review, station history, and direct technical records would still be needed before assigning a phys-

ical explanation. The same caution applies to POC comparisons and public document searches: they characterize available context rather than validate an instrument-change hypothesis.

## 8 Limitations and threats to validity

1. AQS Method Code is reported metadata, not a verified physical hardware, calibration, maintenance, or siting log.
2. A station-local residual can arise from local emissions, meteorology, smoke, network-wide operations, data screening, or other unobserved factors.
3. Geographic donors are selected from pre-event agreement and may not remain valid after an anchor; physical-site deduplication does not remove all reference-series uncertainty.
4. Complete comparisons apply only to the subset with sufficient qualifying distinct donors and retained input windows. The full event ledger documents this selection boundary but cannot eliminate it.
5. Fixed-weight intervals under-cover and split-conformal intervals over-cover on the frozen synthetic evaluation. Neither supports a generic real-event calibration claim.
6. Selection-aware nested intervals model only the fixed observed candidate pool. The frozen full selection-aware synthetic coverage study was infeasible within the deadline, so those intervals are not claimed to have calibrated coverage.
7. Time, donor-as-treated, and date-resampling placebos are diagnostic falsifications. They are not randomized physical experiments or causal probabilities.
8. Same-site POC, QA, and official-document evidence are sparse and contextual. In particular, POC is not a universal physical-instrument identifier, and an unlocated public notice is not evidence of no change.
9. The separate 88502 result is sparse and is not pooled with 88101.
10. The frozen data snapshot, monitoring network, pollutant parameter, time period, and perturbation model limit generalization to other settings.

The Method Code taxonomy is intentionally excluded from these results pending independent human review. Any future stratified analysis must preserve its frozen metadata-only rule, be labelled descriptive, and must not use the already viewed evaluation set to select a new model or decision threshold.

## 9 Conclusion

MetaShift-Bench is a reproducible protocol for auditing reported PM<sub>2.5</sub> measurement-method transitions with metadata anchors, distinct physical donor controls, stable-window synthetic truth, and complete real-event accounting. Its evidence supports a benchmark-and-audit contribution: it records where cross-site diagnostics are available, where they are not, and what the available diagnostics can and cannot say.

The frozen results do not establish a confidence-supported aggregate MetaShift advantage over standard synthetic control. They also do not establish that a reported Method Code transition represents a physical instrument change or a measurement bias. The appropriate use is therefore to prioritize a limited set of candidate station-specific discontinuities for additional records-based and technical review, while preserving unsupported and inconclusive anchors in the public audit trail.

**Table 10:** Frozen evidence release verification record.

| Record                              | Value                 |
|-------------------------------------|-----------------------|
| Evidence tag                        | v0.3.2-evidence-final |
| Frozen evidence commit              | 57d678ecabeb          |
| Synthetic result label              | stable_full_v2        |
| Case manifest SHA-256               | 065b1b65c231c529...   |
| Release-gate checks passed          | 35/35                 |
| Document-consistency checks passed  | 12/12                 |
| Markdown number checks passed       | 57/57                 |
| Two-environment core hashes matched | 34                    |

*Note.* The two-environment statement applies only to the listed designated core artifacts in the frozen reproducibility comparison, not to every local file.

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## 10 Acknowledgements, contribution statement, and required disclosures

**HUMAN COMPLETION REQUIRED.** This section is an intentionally uncompleted truthful-disclosure template. It must be replaced by a 500–1500 word account, verified by every student author and supervising teacher, before the report is submitted. Do not state that a person performed work they did not perform. Do not state that an advisor, school, organization, or AI system did or did not contribute unless the authors have verified the statement.

### Topic origin

**HUMAN COMPLETION REQUIRED:** Explain how the team selected the question, including the intellectual path from the original observation or reading to the metadata-anchor audit formulation. State whether a supervising teacher or other person suggested the topic and how the students refined the question.

### Data acquisition and analysis

**HUMAN COMPLETION REQUIRED:** Identify who downloaded or reconstructed the public EPA AirData inputs, who inspected data rules, who implemented or reviewed the cleaning and donor-uniqueness logic, and who ran or checked the frozen analyses. State only the actual use of public AQS bulk files, public API endpoints, and local credentials. Do not include credentials, raw API responses, or private information.

### Experimental design and writing

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